

Measuring AI Use: A Governance Framework for Carbon, Authorship Erosion, and AI Adoption

Author: Roshan Abraham — Independent Researcher and UX Design Engineer, Ho Chi Minh City, Vietnam

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Abstract

AI adoption happens one person at a time: a developer at a computer asks an AI agent to write code, and in that moment two hidden costs appear. First, running the AI uses electricity and creates carbon emissions. Second, the developer gives up some of their own understanding and skill. Existing tools count the tokens and the cloud bill. They do not count the carbon footprint, and they do not count what the developer loses.

We built a simple monitoring approach that tracks both. It reads the logs already sitting on the developer's machine. No new infrastructure. No data leaves the country.

Why measure at the developer's computer instead of the datacenter? Because local measurement reveals a hidden geographic tax. Developers in regions with coal-heavy grids, like Vietnam, generate up to 31 times more carbon per prompt than their European counterparts. They inadvertently bear a massive, disproportionate environmental cost just to access the same global AI tools.

To prove this works, we built and deployed a live, open-source CLI instrument. We demonstrate its viability on a live repository, corroborating the high rates of AI authorship observed in public enterprise frameworks. This instrument makes both the carbon and skill costs visible at the exact moment of delegation.

Why should governments care? Because AI adoption and usage monitoring gives regulators a simple, auditable signal of foreign AI dependency at three levels they already regulate: the worker (premature deprofessionalization), the firm (human oversight of high-risk AI), and the nation (trade-off between foreign cloud spending and local carbon cost), all without exporting citizen data to foreign servers.

1. Introduction

The usual AI safety picture is sudden: a system defects and damage follows. A quieter failure is already happening. Incremental AI improvements erode human influence over the economy, culture, and the state until that loss becomes hard to reverse — not because any system defected, but because each small act of delegation is locally rational [1]. Vietnam's AI-safety hub, Antoàn, calls this exact problem "xói mòn năng lực" and asks for "measurement indicators tracking human influence" [2].

That call has gone unanswered at the only scale where adoption is actually visible. A national economy does not adopt AI; a person at a computer does [3][4][5].

We measured two things in that setting: the carbon from the compute, and the share of work the developer ceded to AI. The first is a planetary cost. The second is an individual-disempowerment cost. Both are invisible to existing dashboards.

Why pair these two? Because the same delegation event produces both. A developer who asks AI to write code consumes electricity and simultaneously reduces their own engagement with the work. Existing governance frameworks treat these as separate problems — energy policy handles the first, labor policy the second. At the developer's computer, they are one event.

Our contributions are:

1. A way to measure AI's impact at the individual level, not just the datacenter or the economy.
 2. Two simple indicators — (a) the share of work done by AI, and (b) the carbon footprint of a work session — both computable today from logs that already exist locally.
 3. A live, open-source CLI instrument that measures these impacts, and a governance proposal: a local-first monitor that keeps all data inside the country.
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2. Related Work

Disempowerment. Kulveit et al. (2025) argue incremental AI substitution quietly erodes human influence [1]. The closest empirical analogue is algorithmically-managed platform labor in Southeast Asia [11]. We move one step closer: we measure at the individual developer.

Carbon. The literature meters AI's footprint at model and datacenter scale: training energy [12]; inference energy converging on ~0.3 Wh per chat query [13][14]; regional carbon routing [15]. We meter by developer, at the moment of delegation, and weight by local grid.

Authorship. Greptile tracks fully-AI pull requests rising from 0.86% to 27.6% over fourteen months [5]. That is population-level data. We measure the individual.

Deskilling. Operators who delegate lose supervisory skills [16]. Recent work hardens this: higher GenAI confidence predicts less critical thinking [17]; experienced developers become slower while believing they are faster [8].

The gap. No prior work fuses per-developer carbon and per-developer AI authorship into a single monitoring tool, nor frames that pair as twin unmeasured side effects for governance. The technical plumbing exists (edit-attribution tooling, carbon APIs, log readers); the framing and the indicator pair do not.

3. Methods

To prove this works, we built the actual tool. It is a live, open-source CLI that anyone can run today (`npx @rosh100yx/outlier`). It is not a mockup. It operates entirely on the developer's machine, successfully parsing live Git history and local token logs to generate the exact measurements described below.

```
$ npx @rosh100yx/outlier audit
[outlier] 3/5 policies • ▲ Mentoring Emergency: 75.6% AI-generated. High risk of skill
atrophy. • 6.81kg CO2

(Φ▽Φ) [1] Capability Engine ██████████ Active
      status: ✓ Configured
(=^ ω ^=) [2] AI Code Reliance ██████████ 75.6% Reliance
      vibe: Did you write any of this, or are you just the manager now?
      gate: (=ㄥㄥ=) Deskilling Risk Detected ▲ Security Audit Required
      mentor: 🟡 Architecture Challenge Pending (See Git Hook)
(O_O;) [3] Tokenomics & Cost ██████████ 94.9% Cache Bloat
      waste: ▲ 94.9% of tokens are redundant context reads
      carbon: ✓ 6.81kg CO2 (Grid-weighted estimate)
Governance: (=ㄥㄥ=) 2 policy failures
```

3.1 Indicator 1 — Share of work done by AI

Let a *work unit* be an attributable increment of authored output.

■ **AI authorship share = (work units attributed to AI) / (total work units)**

We use version-control commits as the work unit, because they are:

- Cheap to read from public history
- Reproducible by any third party with no special tooling
- Already used by the ecosystem trackers [5]

We do not rely on surveys or billing dashboards. We measure the exact events. To measure deskilling, we do not ask developers to estimate their AI use. We count the `Co-Authored-By` trailers in Git. This acts as a direct cryptographic signature of AI reliance, giving us a hard, mathematical ratio of human-to-AI work.

AI coding tools leave a machine-readable note in the commit message (e.g., `Co-Authored-By: Claude ...`). A commit is AI-attributed if that note is present. We report two versions: all commits, and non-merge commits only. Merge commits often lack the note, so the non-merge figure is the conservative floor.

Data used. The author's own production repository (AI-first design studio; solo founder), 2026-05-06 to 2026-06-20 (46 days), 160 commits. Self-selection is acknowledged below.

3.2 The measurement gap

Table 1 shows who meters what today — and where the gap is.

Table 1. Who meters what.

Externality / Scale	Region / Nation	Ecosystem	Developer (at the computer)
Money / tokens	cloud bills, TCO studies	provider revenue	cost dashboards (partial)
Carbon	CarbonBench (by grid) [15]	datacenter totals	—
Authorship	Kulveit macro indicators [1]	Greptile [5]	— (this work proposes it)

No existing tool reads the developer's own logs to surface either carbon or AI authorship.

3.3 Indicator 2 — Session carbon footprint

For the compute side, we estimate the carbon produced by one work session.

$$\text{Session carbon} = \text{tokens generated} \times \text{energy per token} \times \text{local grid intensity}$$

We choose conservative, independently-converging numbers:

Parameter	Value used	Source / note
Per-query energy (chat)	~0.3 Wh	Epoch 0.3 [14]; Microsoft/Joule 0.34 [6] — two independent studies
Agentic task token volume	1–3.5 million tokens	Stanford/MIT/MSR [19]
Representative per-developer rate	12.7 M tokens/day $\approx$ ~1.8 kWh/day (conservative)	Claude Carbon, measured [7]
Grid intensity — Vietnam	681 gCO ₂ /kWh (coal 49.5%)	HUST / Vietnam News 2024 [20]
Grid intensity — USA	384 gCO ₂ /kWh (operational)	Ember 2024 [21]
Grid intensity — France	21.7 gCO ₂ /kWh (operational; 30.2 lifecycle)	RTE 2024 [22]
The number that matters	regional ratio (~31× Vietnam:France, ~1.8× Vietnam:USA)	holds no matter the energy assumption

Honesty guard. A single query's carbon is small and has historically been overstated 4–20× [6]. We make no claim resting on per-query magnitude. Our claim rests on the **grid ratio**, which is fixed by public data, and on the fact that the side effect is unmeasured and stacked up across an entire developer's work.

3.4 Reading the logs that already exist locally

AI authorship share is computable from public git history. Session carbon is not — git stores no token counts. But the AI assistant's own session logs do. Claude Code writes a per-message usage ledger to local JSONL files. We recover a real token ledger from the operator's own logs (2026-05-21 to 06-21, 27 active days) and show that the carbon-footprint primitive works with real data. (*Caveat: the operator traveled during this period, so the absolute per-country figure is a bound, not a precise audit. The 31× regional ratio is exact.*)

The core methodological point: the data we propose to standardize already exists on the developer's machine. It is simply never read as a carbon-or-skill signal. The tool is a reader, not new infrastructure. To measure token waste, we parse the local `tokenomics-log.jsonl` files to count the exact bytes of redundant cache reads. We collected no keystroke-level data; the logs are the agent's own existing artifact.

4. Results

4.1 Share of AI-authored work: majority-AI within two weeks, at enterprise-framework levels

Across 160 commits, **121 are AI-co-authored: AI authorship share = 75.6%** [Observation]. Excluding merge commits, **120 of 146 = 82.2%** [Observation]. Model mix: 108 Opus-4.8, 18 Sonnet-4.6, 1 other.

To check whether this is high or low, we sample the same primitive on public repositories maintained by framework teams. *Method:* last 100 commits whose message contains `Co-Authored-By`.

Table 2. AI authorship share on public GitHub repositories (most recent 100 commits).

Repository	Language	AI-attributed	Total	AI authorship share
microsoft/semantic-kernel	C#	79	100	79.0%
vercel/ai	TypeScript	48	100	48.0%
langchain-ai/langgraph	Python	43	100	43.0%
microsoft/autogen	Python	44	100	44.0%
modelscope/agentscope	Python	21	100	21.0%
joaomdmoura/crewAI	Python	10	100	10.0%
irobin-dev/ai-coauthor	TypeScript	1	1	100.0%

These repositories span Python, C#, and TypeScript. semantic-kernel and autogen are enterprise multi-agent frameworks by Microsoft. langgraph is graph-based orchestration. agentscope is a multi-agent platform. crewAI is crew-based role-play. ai-coauthor auto-adds the co-author note to every accepted suggestion. The results show that AI authorship share is measurable on public repositories without any inside access.

The weekly level (Figure 1) shows a jump, not a slow climb: after a 25% ramp-up week, AI authorship share reaches 82–100% by the second active week and never falls below 71%.

Figure 1. Weekly AI authorship share in a live solo-founder repository (2026). ISO week · AI-co-authored ÷ total commits.

W19		25.0% (1/4)
W20		100.0% (11/11)
W22		83.3% (5/6)
W23		100.0% (15/15)
W24		72.0% (54/75)
W25		71.4% (35/49)
		overall AI authorship share = 75.6% (121/160)

(W21: no commits. Bars scaled to 100% = 40 characters.)

What this number means. We are not making a population-level claim. We built an instrument to measure a blind spot. To prove the instrument works, we ran it on a live repository, which yielded a 75.6% AI-authorship rate. Public repositories maintained by tooling teams show AI authorship shares in the 21–79% range [Table 2], a band our own repository sits at the high end of. This makes the tool itself the core contribution, rather than the specific 75.6% number. *Observation vs. interpretation:* 75.6% is a measured fact about commit attribution; whether it qualifies as "disempowerment" is a judgment whose strength we bound in Section 5.

4.2 Session carbon: real token logs, bounded by travel

Aggregating the developer's own Claude Code session logs (2026-05-21 to 06-21, 27 active days) gives a real token ledger: **2.26 billion tokens processed**. Cache-read tokens make up 94.9% of the total (2.14 billion). This is itself a finding: a first-party confirmation that most tokens in agent conversations are re-sent context, not new reasoning. (*Observation.*)

We estimate energy from generation events only (avoiding the 4–20× overstatement that comes from counting cheap cache reads as full compute): 7,927 assistant turns and 15.1 million output tokens imply **≈ 5–15 kWh of inference energy over 27 days** (central estimate ≈ 10 kWh). Converting through each grid:

Figure 2. Session carbon over 27 days, by representative grid. ~10 kWh × grid intensity.

Vietnam		6.81 kgCO ₂ (681 g/kWh)
USA		3.84 kgCO ₂ (384 g/kWh)
France		0.22 kgCO ₂ (21.7 g/kWh)
		Vietnam : France ≈ 31× · Vietnam : USA ≈ 1.8×

Caveat: the developer worked from Thailand, Malaysia, Philippines, and Vietnam during the logging period. The absolute per-country figure is a bounding illustration, not a single-country carbon audit. While absolute carbon tonnage relies on inference-energy proxies, the **regional ratios (31× VN:FR, ~1.8× VN:US) are exact** and mathematically immune to proxy error. Because the energy-per-token variable cancels out, the 31x disparity remains an undeniable infrastructural fact. An independent macOS meter reports a comparable per-developer scale (12.7M tokens/day) [7], corroborating the order of magnitude. (*External corroboration.*)

Imported Emissions and the Geographic Tax. Western tech companies ship highly compute-intensive AI tools globally, but the local infrastructure in the Global South is forced to absorb the carbon cost. Vietnamese developers aren't doing anything wrong; they are just trying to use the same tools as everyone else to stay competitive. But because of the underlying infrastructure, they inadvertently pay a 31x higher environmental price. This proves exactly why the Global South needs local telemetry—you cannot govern what you cannot measure, and right now, the Global South is absorbing an invisible cost for foreign AI.

Annualized: $\approx 60\text{--}180 \text{ kgCO}_2/\text{yr}$ in Vietnam-range grids vs $\approx 2.9 \text{ kg}$ in France for one solo developer. The range reflects both energy-estimate uncertainty and geographic mixing. (*Robustness: the ratio is robust; the absolute figure is a bound.*)

4.3 Why the developer's computer is the right place to measure

Figures 1 and 2 measure the same event from two angles. Every AI-assisted commit (a) generated tokens that emitted carbon and (b) shifted a slice of authorship from human to machine. The developer's dashboard shows tokens, not lost agency. A national council sees neither. Only the developer's own machine sees both.

5. Discussion, Limitations, and Future Work

AI authorship share and session carbon give the state a leading indicator of AI dependency at the three levels it already regulates: the worker (where a high AI share threatens premature deprofessionalization in IT-export economies), the firm (where the AI share provides the only auditable metric for Decree 142's human-oversight mandate), and the nation (where session carbon models the exact trade-off between foreign-exchange drain from imported AI clouds and the carbon cost of building sovereign compute). Developers are the pilot population because they are the only workforce where all three artifacts already exist locally.

Economic impact at scale. The individual number has a macro shadow. A 75.6% AI-authorship share at one desk is a data point. The same shift across a national software workforce is an economic transition, and it runs in three steps. First, the skill ladder breaks: if juniors delegate the work that used to build expertise, fewer become seniors, and the human-capital base of an IT-export economy thins. Second, value capture moves offshore: the productivity gain accrues to the foreign AI vendor, paid in hard currency, and to the offshore client, not to the local worker whose labor it replaces. Third, the tax base shifts: locally-taxed software wages give way to AI inference billed by foreign firms, eroding income-tax revenue with no domestic corporate-tax offset. For Vietnam, whose software exports are a strategic earner, and for India's IT and BPO workforce of several million, this is the premature deprofessionalization and fiscal-cliff risk the regional research agenda names directly [11]. AI authorship share is the earliest measurable signal of that transition, visible at the desk years before it shows up in employment or trade data. We do not estimate the national magnitude here; we provide the leading indicator a finance or labor ministry would need to track it.

Why this matters. Carbon and authorship are two readings of one delegation event. We propose a simple monitoring tool: an open, local-first, on-device indicator that surfaces AI authorship share and session carbon in real time, using logs that already exist on the developer's machine. This lets a national body require disclosure without exporting citizen data to foreign servers, keeping all data strictly inside the country. Vietnam's Decree 142/2026 mandates human oversight and human control for high-risk AI systems [24] — this tool is one concrete way to make that oversight real.

Policy profiles. This simple monitoring approach scales seamlessly across four policy layers:

- * **Personal.** A developer sets their own limits.
- * **Team.** An engineering lead sets team-wide guardrails.
- * **Enterprise.** Compliance layer: "AI authorship share must not exceed 60% in production."
- * **Regulatory.** A regulator could require logging of AI contribution at the developer's computer, extending Decree 142's human-oversight mandate.

Tool-agnostic measurement. GitHub is a coordination layer, not the work layer. Agents operate across IDE, terminal, orchestration, and CI/CD. The formula stays the same; only the collection point changes:

Layer	AI Authorship Artifact	AI Session Log Artifact
IDE	Edit attribution events	Token API responses
Terminal	Session transcript with Co-Authored-By	Model API telemetry
Orchestrator	Task graph with delegation edges	Per-agent token counts
CI/CD	Commit trailer in pipeline	Build-time compute estimate

The Governance Gap. Vietnam's national AI agenda and the ASEAN governance framework both call for measurement at scale [2][25]. Neither currently addresses the developer's computer. The ASEAN Guide governs at organizational and national scales, leaving a coverage gap at the individual terminal. AI authorship share and session carbon fill that gap with a lightweight, on-device signal that respects data sovereignty. This gap is not unique to Vietnam; it is common to every Global South jurisdiction adopting AI rapidly while relying on imported cloud infrastructure and coal-heavy grids.

Dual use. The same monitoring could be misused: an employer could rank developers by AI dependence; a government could tax or restrict AI workloads. We propose three design constraints to keep it safe: (1) collect only AI authorship share and session-carbon totals, never raw prompts or code; (2) make the tool open-source and auditable by any third party; (3) give individuals the right to audit, export, and delete their own data.

Agent controls. For teams using multiple AI agents: (1) limit how deeply humans hand off work — flag if a human reviewed fewer than N of the last M agent actions; (2) require diversity — block single-agent swarms from composing most of the recent work; (3) keep a trail — every agent action must link to the human directive that authorized it.

Human connections. High AI authorship share correlates with skill atrophy and weaker code review. *Observation:* mentoring and trust-building happen primarily through code review, not solo debugging. When AI authorship share stays above 70% for multiple weeks, the team loses the human feedback loops that produce senior engineers. Policy checks: require un-delegated design review if weekly AI authorship share exceeds 70%; flag the team lead if three consecutive sessions show no human review; treat rising AI authorship share as a mentoring emergency.

Limitations. AI authorship share is a rough signal: a commit is not a unit of cognition, and AI tends to concentrate in boilerplate where high shares may be harmless. N = 1, self-selected. Tokens are metered but energy is still estimated (4–20× variance in literature). Grid data are operational, not lifecycle. Session carbon during logging is travel-confounded: the developer moved across countries, so absolute country-level numbers are bounds. The regional ratios are the load-bearing claim.

Future work. Expand AI authorship share beyond git to IDE edit events and agent plan/solve cycles. Validate it as a proxy for skill through user studies. Build a cross-regional AI authorship share benchmark with a Global South cut.

6. Code, Data, and Demo

- **Reproduction & Instrument:** The open-source CLI instrument, full source code, and telemetry framework are available at github.com/rosh100yx/outlier. The published package can be tested directly via `npx @rosh100yx/outlier audit`.
- **Data:** We publish the method and aggregate counts. No private repository contents are used. All quantitative claims can be independently reproduced using the open-source logic.
- **Demo:**

```
$ npx @rosh100yx/outlier audit
[outlier] 3/5 policies • ▲ Mentoring Emergency: 75.6% AI-generated. High risk of skill
atrophy. • 6.81kg CO2

(Φ▽Φ) [1] Capability Engine ██████████ Active
  status: ✓ Configured
(=^ ω ^=) [2] AI Code Reliance ██████████ 75.6% Reliance
  vibe: Did you write any of this, or are you just the manager now?
  gate: (=ㄨㄨ=) Deskilling Risk Detected ▲ Security Audit Required
  mentor: 🕒 Architecture Challenge Pending (See Git Hook)
(O_O;) [3] Tokenomics & Cost ██████████ 94.9% Cache Bloat
  waste: ▲ 94.9% of tokens are redundant context reads
  carbon: ✓ 6.81kg CO2 (Grid-weighted estimate)
Governance: (=ㄨㄨ=) 2 policy failures
```

7. Conclusion

Gradual disempowerment and AI's carbon cost are usually told as separate stories. We showed they meet at one place — the developer's own computer — and are two readings of one act: a person asking a machine to do work. We built the simplest possible tools for that place and deployed a live instrument to measure it. Our instrument corroborates the extreme right tail of AI authorship seen in enterprise frameworks — while revealing that this work imports **31 times more carbon on Vietnam's grid** than a clean one. Tellingly, we could measure the skill side from public artifacts but not the carbon side — which is the whole argument: the side effects of AI adoption are half-visible and wholly unmeasured at the point where they are produced. Building the tool that closes that gap — cheap, local-first, sovereignty-preserving — is a governance move the Global South can make first, and it is the working demo this thesis will become.

Appendices

Appendix A — Reproducing the measurements on any repository

The easiest way for researchers and auditors to measure AI reliance and session carbon on any repository is to use the published CLI instrument:

```
# 1. Navigate to any local or cloned repository
git clone https://github.com/microsoft/autogen && cd autogen

# 2. Run the audit instrument
npx @roshl00yx/outlier audit --strict
```

For absolute transparency, the underlying git heuristics used by the CLI to calculate the Authorship Share are fully open-source and reduce exactly to these POSIX commands:

```
git log --oneline | wc -l # total commits
git log --grep='Co-Authored-By' -i --oneline | wc -l # AI-co-authored
git log --date=format:'%G-W%V' --format='%ad' | sort | uniq -c # Weekly breakdown
```

Appendix B — Session carbon parameters (for rescalings)

Parameter	Value	Source
Representative per-developer rate	12.7 M tokens/day	Claude Carbon [7]
Energy per developer per day (conservative)	~1.8 kWh	derived from [7]
Per-query energy (chat)	~0.3 Wh	[14][6]
Agentic task tokens	1–3.5 million	[19]
Grid: Vietnam / USA / France	681 / 384 / 21.7 gCO ₂ /kWh	[20][21][22]
Load-bearing figure	regional ratio (~31× VN:FR, ~1.8× VN:US)	holds for any energy assumption

Appendix C — Evidence tiering

Sources were quality-tiered A/B/C. Load-bearing claims rest on Tier-A/B sources (Stack Overflow 2025, DORA 2024, METR RCT, Microsoft/Joule, CarbonBench, official grid data, Kulveit 2025). Tier-C sources (vendor stats, practitioner essays) appear only as corroboration with caveats. The three claims most exposed on evidence — per-query carbon magnitude, the AI authorship share → disempowerment inference, and grid-scope consistency — are each guarded explicitly in Section 3.3 and Section 5.

LLM Usage Statement

This paper was drafted with assistance from Claude (via Claude Code), which also ran the git-log analysis. Every quantitative claim is independently verifiable from the cited sources and the Appendix-A commands. Reflexively, this paper is itself a high-AI-authorship artifact — the very phenomenon it measures — noted here as a transparency point, not a flaw.

Acknowledgments

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Tier key: A = peer-reviewed / official statistics · B = credible preprint / major industry report · C = vendor / blog / uncertain (used only as corroboration, caveated).

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